CSS

Outline, table, list, inline, block , inline-block , flex

Theory Notes

1. CSS Flexbox

Definition:

Flexbox (Flexible Box Layout) is a one-dimensional layout model in CSS that allows you to design layouts that align and distribute space efficiently along a single axis(row or column) — even when the size of the items is unknown or dynamic. It's best for arranging elements in a line, such as navbars, menus, or card lists.

Basic Syntax:

```
.container {

display: flex; /* Enables flexbox */

flex-direction: row; /* row | column | row-reverse | column-reverse */

justify-content: center; /* main axis alignment */

align-items: center; /* cross axis alignment */

flex-wrap: wrap; /* wrap items onto multiple lines */

gap: 20px; /* space between items */

}
```

1. Tables

- <table> → defines a table.
- <tr> → table row.
- <th> → table header cell.
- <td> → table data cell.
- CSS properties:
    - border → adds borders around cells.
    - border-collapse: collapse; → merges adjacent borders.
    - padding → adds space inside cells.
    - border-spacing → controls space between cells (when collapse is not used).

2. Lists

- **Ordered list (**<ol>**)** → numbered items.
  - list-style-type → controls numbering style (decimal, upper-alpha, lower-roman, lower-greek).
- **Unordered list (**<ul>**)** → bulleted items.
  - list-style-type → controls bullet style (disc, circle, square).
  - list-style-image → replaces bullets with an image.
  - list-style-position → inside or outside.

3. Display Properties

- **Block elements** → take full width, start on a new line (<p>,).
- **Inline elements** → flow with text, ignore width/height (, <a>).
- **inline-block** → behaves like inline but respects width/height.

4. Flexbox

- **Parent container** → display: flex;.
- Properties:
  - flex-direction → row (default) or column.
  - justify-content → alignment along main axis (center, space-between, space-around).
  - align-items → alignment along cross axis (center, flex-start, flex-end).
  - gap → spacing between flex items.
  - flex-wrap → allows items to wrap onto multiple lines.

5. Outline vs Border

- **Border** → part of the box model, affects layout.
- **Outline** → drawn outside the border, does not affect layout.
- Example: outline: 2px solid red;.

Practice Tasks

Short Answer

1. What is the difference between border and outline?
2. How does border-collapse: collapse affect table styling?
3. What is the difference between inline and inline-block?
4. How does justify-content: center differ from align-items: center in flexbox?

Coding Exercises

1. Create a table with 3 rows and 3 columns, styled with border-collapse: collapse and padding: 10px.
2. Make an ordered list with Roman numerals and an unordered list with custom image bullets.
3. Demonstrate the difference between inline, block, and inline-block using <p> and .

4. Style an input field with a thick outline and compare it to a border.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Table</title>
    <style>
      input {
        /* outline-width: 10px;
            outline-style: solid;
            outline-color: brown; */
        /* outline: 10px solid brown ; */
        margin-top: 20px;
      }

      /* div {
        width: 200px;
        height: 200px;
        background-color: gray;
        color: white;
        overflow-y: scroll;
      } */
      table, td, th {
        border: 2px solid black;
        border-collapse: collapse;
        padding: 10px;
        /* border-spacing: 30px; */
      }
      ul{

        /* list-style-type: circle; */
        /* list-style-position: inside; */
        /* list-style-image: url("https://lh3.googleusercontent.com/-
4GxJOlrZu60/AAAAAAAAAAI/AAAAAAAAAAA/ALKGfklJPNSk3uYOkO5E44OUgCm080GZPA/photo.j
pg?sz=46"); */
        list-style: url("https://lh3.googleusercontent.com/-
4GxJOlrZu60/AAAAAAAAAAI/AAAAAAAAAAA/ALKGfklJPNSk3uYOkO5E44OUgCm080GZPA/photo.j
pg?sz=46") inside;
      }
      ul>li{
        text-transform: uppercase;
        border: 2px solid black;
      }

      ol{
        list-style-type: lower-greek;
        /* upper-alpha,
```

```
        lower-alpha,
        upper-roman,
        lower-roman, */
        /* list-style-image: url("https://lh3.googleusercontent.com/-
4GxJOlrZu60/AAAAAAAAAAI/AAAAAAAAAAA/ALKGfklJPNSk3uYOkO5E44OUgCm080GZPA/photo.j
pg?sz=46"); */

      }
      p,span{
        display: inline-block;
        width: 100px;
        height: 100px;
        /* inline -> width & height ignore */
      }
      /* span{
        width: 200px;
        height: 200px;
      } */

      .parent{
        height: 700px;
        border: 2px solid black ;
        display: flex;

        /* flex-direction: column; */
        gap: 20px;
        justify-content: center;
        align-items: center;
        /* flex-wrap: wrap; */
      }
      .child{
        width: 100px;
        height: 150px;
        background-color: brown;
        color: white;
        /* margin-left: 20px; */
      }
    </style>
  </head>
  <body>
    <section class="parent">
        <div class="child">Box 1</div>
        <div class="child">Box 2</div>
        <div class="child">Box 3</div>

    </section>
```

```html
    <p>block element</p>
    <span>inline element</span>
    <span>inline element</span>
    <p>block element</p>

    <ol>
        <li>html</li>
        <li>css</li>
        <li>javascript</li>
    </ol>
    <ul >
        <li>html</li>
        <li>css</li>
        <li>javascript</li>
    </ul>

    <table >
       <tr>
        <th>Name</th>
        <th>Age</th>
        <th>Degree</th>
       </tr>

       <tr>
        <td>Ganesh</td>
        <td>24</td>
        <td>LLB</td>
       </tr>

    </table>


    <h1>Outline</h1>
    <div>
      voluptate minus quibusdam eos. Repudiandae, libero? Facilis ducimus
      eveniet ipsum omnis dicta, eum eius possimus alias, explicabo cumque
      praesentium inventore, consequuntur dolor. Voluptas corporis obcaecati
    </div>

    <input type="text" placeholder="Enter your name " />
  <!-- </body> -->
</html>
```